

CE 49X: Introduction to Computational Thinking and Data Science
Midterm Exam – Part 1 (Paper-Based)
Weeks 1–7 Material

Name: _____ **Student ID:** _____

Date: April 8, 2026

Duration: 60 minutes

Instructions:

- This is Part 1 of the midterm exam (paper-based).
 - Answer all questions. Show your work for full credit on written questions.
 - One page (both sides) of handwritten notes is allowed.
 - No calculators or electronic devices.
 - A helpful formulas page is provided at the end.
 - Total points: **50**
-

Section A: Multiple Choice

[20 points]

Circle the single best answer for each question. Each question is worth 2 points.

1. What does the following expression evaluate to?

```
len([x for x in range(10) if x % 3 == 0])
```

- (a) 3
- (b) 4
- (c) 10
- (d) 0

2. Which import statement allows you to call `np.array([1, 2, 3])`?

- (a) `import numpy`
- (b) `from numpy import array`
- (c) `import numpy as np`
- (d) `import np from numpy`

3. Given `a = np.array([1, 2, 3, 4, 5])`, what does `a[1:4]` return?

- (a) `array([1, 2, 3, 4])`
- (b) `array([2, 3, 4])`
- (c) `array([2, 3, 4, 5])`
- (d) `array([1, 2, 3])`

4. Which pandas method fills missing values with the **previous** row's value?

- (a) `df.dropna()`
- (b) `df.fillna(method='ffill')`
- (c) `df.interpolate()`
- (d) `df.fillna(0)`

5. Which visual encoding channel is **most accurate** for comparing quantitative values?

- (a) Color intensity
- (b) Angle
- (c) Area
- (d) Position

- 6.** According to Tufte’s principles, what should a well-designed chart maximize?
- (a) Number of colors used
 - (b) Data-ink ratio
 - (c) Chart area
 - (d) Number of gridlines
- 7.** A dataset has mean $\mu = 50$ and standard deviation $\sigma = 8$. What is the z-score of a value of 66?
- (a) 1.5
 - (b) 2.0
 - (c) 2.5
 - (d) 8.0
- 8.** A hypothesis test yields a p-value of 0.03 at significance level $\alpha = 0.05$. What is the correct conclusion?
- (a) There is a 3% chance that H_0 is true
 - (b) Reject H_0
 - (c) Fail to reject H_0
 - (d) The effect size is 0.03
- 9.** A model predicts “safe” for every sample in a dataset where 95% of samples are actually safe. Its accuracy is 95%. Why is this misleading?
- (a) The model is overfitting
 - (b) It never identifies the dangerous 5%
 - (c) The training set is too small
 - (d) Accuracy requires cross-validation
- 10.** In Naive Bayes, what does the “naive” assumption mean?
- (a) The model uses no prior information
 - (b) Features are conditionally independent given the class
 - (c) The model always predicts the most common class
 - (d) Training data must be normally distributed

Section B: Written Questions**[30 points]****Question 1: Code Tracing & Debugging****[6 points]**

(a) (3 points) What is the **exact** output of the following code?

```
beams = [3.5, 7.2, 4.8, 12.1, 6.3]
long_beams = [b for b in beams if b > 5.0]
total = sum(long_beams)
count = len(long_beams)
print(f"Found {count} beams totaling {total:.1f} m")
print(f"Average: {total/count:.2f} m")
```

(b) (3 points) The following function has a bug. Identify the error, explain what the code currently prints, and write the corrected line(s).

```
def classify_soil(spt_n_values):
    """Classify as 'soft' (N<10), 'medium' (10<=N<30), 'hard' (N>=30)"""
    results = {}
    for n in spt_n_values:
        if n < 10:
            results[n] = 'soft'
        elif n < 30:
            results[n] = 'medium'
        else:
            results[n] = 'hard'
    return results

data = [5, 25, 5, 40, 25]
print(len(classify_soil(data)))
```

Question 2: NumPy & Pandas**[6 points]**

(a) (3 points) Explain the difference between a NumPy **view** and a **copy**. Given the code below, what is the value of `speeds[1]` after execution? Explain why.

```
import numpy as np
speeds = np.array([40, 55, 30, 65, 20])
subset = speeds[1:4]
```

```
subset[0] = 99
```

(b) (3 points) Write a **single** pandas expression that filters a DataFrame `df` to rows where `AVERAGE_SPEED < 20 AND NUMBER_OF_VEHICLES > 100`, and returns only the `GEOHASH` and `AVERAGE_SPEED` columns.

Question 3: Visualization**[6 points]**

(a) **(3 points)** A colleague prepares a chart for a municipal earthquake damage report. The chart is a **3D pie chart** using the **jet (rainbow) colormap** to show the percentage of damaged buildings across 8 districts. Identify **two** specific problems with this chart and suggest a fix for each.

(b) **(3 points)** What is **Anscombe's Quartet**? In 2–3 sentences, explain what it demonstrates and the lesson it teaches for data analysis.

Question 4: Statistical Analysis**[6 points]**

(a) **(3 points)** A batch of 28-day concrete cylinder tests gives a mean compressive strength of $\mu = 32.5$ MPa with a standard deviation of $\sigma = 3.2$ MPa. Compute the z-score for a cylinder that tested at 26.1 MPa. Is this result “extreme” by the $|z| > 2$ convention?

(b) **(3 points)** Define **Type I error** and **Type II error**. For a concrete strength test where H_0 : “the concrete meets the 30 MPa specification,” describe what each error type means in practical engineering terms. Which is more dangerous for structural safety?

Question 5: Machine Learning & Naive Bayes**[6 points]**

(a) (3 points) A soil liquefaction classifier is evaluated on 200 test sites. The confusion matrix is:

	Predicted Stable	Predicted Liquefiable
Actually Stable	160	15
Actually Liquefiable	5	20

Compute the **precision** and **recall** for the “Liquefiable” class. Should the engineer prioritize precision or recall in this application? Justify in 1–2 sentences.

(b) (3 points) A bridge inspector uses Bayes’ theorem to assess beam condition. Given:

- Prior: $P(\text{Good}) = 0.70$, $P(\text{Poor}) = 0.30$
- Likelihood of observing an 8mm crack: $P(8\text{mm} \mid \text{Good}) = 0.02$, $P(8\text{mm} \mid \text{Poor}) = 0.15$

Compute $P(\text{Poor} \mid 8\text{mm crack})$. Show your work.

Helpful Formulas

Statistics:

$$\text{Z-score: } z = \frac{x - \mu}{\sigma}$$

$$68\text{-}95\text{-}99.7 \text{ Rule: } P(\mu \pm 1\sigma) \approx 68\%, \quad P(\mu \pm 2\sigma) \approx 95\%, \quad P(\mu \pm 3\sigma) \approx 99.7\%$$

Classification Metrics:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

$$\text{Precision} = \frac{TP}{TP + FP} \quad \text{Recall} = \frac{TP}{TP + FN}$$

$$F_1 = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

Confusion Matrix Layout:

	Predicted Negative	Predicted Positive
Actually Negative	TN	FP
Actually Positive	FN	TP

Bayes' Theorem:

$$P(A | B) = \frac{P(B | A) \cdot P(A)}{P(B)}$$

$$\text{where } P(B) = \sum_i P(B | A_i) \cdot P(A_i)$$

Naive Bayes:

$$P(L | f_1, \dots, f_n) \propto P(L) \prod_{i=1}^n P(f_i | L)$$

Gaussian PDF:

$$P(f_i | L) = \frac{1}{\sqrt{2\pi\sigma_L^2}} \exp\left(-\frac{(f_i - \mu_L)^2}{2\sigma_L^2}\right)$$

Visual Encoding Accuracy Hierarchy:

Position > Length > Angle > Area > Color Intensity

End of Part 1

Please review your answers if time permits.